семинар 2

WN - стац. процесс

$$\begin{cases} E = 0, \\ Var = \sigma_\varepsilon^2 \\ cov = 0 \end{cases}$$

1

$X_t \sim iid$ $\quad Z_t \sim iid$

$$X_i = \begin{cases} 1, & p = 1/2 \\ -1, & p = 1/2 \end{cases} \qquad Z_i = \begin{cases} 1, & p = 1/2 \\ 0, & p = 1/2 \end{cases}$$

X_t, Z_t независимые

$$y_t = X_t(Z_t + Z_{t-1})$$

$E(Z_t) = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{2}$

① $E(y_t) = E(X_t Z_t) + E(X_t Z_{t-1}) = \underbrace{E(X_t)}_{0} \cdot E(Z_t) + E(X_t) \cdot E(Z_{t-1}) = 0$

$E(X_t) = 1 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$

$Z_t^2 = \begin{cases} 1, & p = 1/2 \\ 0, & p = 1/2 \end{cases}$ (то же самое)

② $Var(y_t) = E(y_t^2) - \underbrace{E^2(y_t)}_{0} = E(X_t^2(Z_t + Z_{t-1})^2) = \underbrace{E(X_t^2)}_{X_t^2 = 1 \Rightarrow E(X_t^2) = 1} \cdot E(Z_t^2 + 2Z_tZ_{t-1} + Z_{t-1}^2) = \underbrace{E(Z_t^2)}_{1/2} + \underbrace{2E(Z_t) \cdot E(Z_{t-1})}_{1/4,\ 1/2} + \underbrace{E(Z_{t-1}^2)}_{1/2} = 3/2$

③ $cov(y_t, y_t) = Var(y_t) = \gamma(0) = 3/2$

$cov(y_t, y_{t-1}) = \gamma(1) = cov(X_tZ_t + X_tZ_{t-1},\ X_{t-1}Z_{t-1} + X_{t-1}Z_{t-2}) = cov(X_tZ_t,\ X_{t-1}Z_{t-1}) + \ldots =$

$cov(a,b) = E(ab) - E(a)E(b)$

$= \underbrace{E(X_tZ_tX_{t-1}Z_{t-1})}_{E(X_t)E(Z_t)\cdot\ldots\ =\ 0} - E(X_tZ_t) \cdot \underbrace{E(X_{t-1})}_{0} E(Z_{t-1}) + \ldots = 0$

$cov(y_t, y_{t-k}) = \gamma(k) = 0$ из-за $E(X_t) = 0$ в каждом слагаемом

y_t это белый шум ($E(y_t) = 0$, $Var(y_t) = const$, $cov(y_t, y_{t-k}) = 0$)

2

$MA(2)$ (кол-во лагов): $y_t = 5 + \varepsilon_t + 3\varepsilon_{t-1} - 2\varepsilon_{t-2}$, $\varepsilon_i : WN(0, \sigma_\varepsilon^2)$

$E(y_t) = 5 + 0 + 3 \cdot 0 - 2 \cdot 0 = 5$

$Var(y_t) = Var(\varepsilon_t) + 9Var(\varepsilon_{t-1}) + 4Var(\varepsilon_{t-2}) + 2 \cdot 3 \underbrace{cov(\varepsilon_t, \varepsilon_{t-1})}_{0} + 2 \cdot (-2) \underbrace{cov(\varepsilon_t, \varepsilon_{t-2})}_{0} + 2 \cdot 3 \cdot (-2) \underbrace{cov(\varepsilon_{t-1}, \varepsilon_{t-2})}_{0 \text{ по св-вам } WN} = 14\sigma^2 = \gamma(0)$

$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\,cov(X,Y)$

$\gamma(0) = cov(y_t, y_t) = Var(y_t) = 14\sigma^2$

$\gamma(1) = cov(y_t, y_{t-1}) = cov(5 + \varepsilon_t + 3\varepsilon_{t-1} - 2\varepsilon_{t-2},\ 5 + \varepsilon_{t-1} + 3\varepsilon_{t-2} - 2\varepsilon_{t-3}) = 3cov(\varepsilon_{t-1}, \varepsilon_{t-1}) - 6cov(\varepsilon_{t-2}, \varepsilon_{t-2}) = 3 \cdot \sigma^2 - 6\sigma^2 = -3\sigma^2$

$\gamma(2) = cov(y_t, y_{t-2}) = cov(5 + \varepsilon_t + 3\varepsilon_{t-1} - 2\varepsilon_{t-2}$

$5 + \varepsilon_{t-2} + 3\varepsilon_{t-3} - 2\varepsilon_{t-4}) = -2cov(\varepsilon_{t-2}, \varepsilon_{t-2}) = -2\sigma^2$

Все следующие ковариации 0 (больше не будет пары для каждого лага) $\Rightarrow \gamma(k) = 0 \quad k \geq 3$

AC - автоковариации

ACF - автокорреляции

$\rho(0) = corr(y_t, y_t) = 1$

$\rho(1) = corr(y_t, y_{t-1}) = \frac{cov(y_t, y_{t-1})}{\sqrt{Var(y_t)} \cdot \sqrt{Var(y_{t-1})}} = \frac{-3\sigma^2}{14\sigma^2} = -\frac{3}{14}$

равны

$\rho(2) = corr(y_t, y_{t-2}) = \frac{cov(y_t, y_{t-2})}{\sqrt{Var(y_t)} \cdot \sqrt{Var(y_{t-1})}} = \frac{-2\sigma^2}{14\sigma^2} = -\frac{2}{14}$

равны

$\rho(3) = \rho(4) = \ldots = \rho(k) = 0$

Процесс слабо стационарен: ничего от времени не зависит.

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3 AR(1): $y_t = 2 + 0.5 y_{t-1} + \varepsilon_t$, $\varepsilon_i : WN(0, \sigma_\varepsilon^2)$

Пусть процесс стационарен.

$E(y_t) = E(2 + 0.5 y_{t-1} + \varepsilon_t) = E(2) + 0.5E(y_{t-1}) + E(\varepsilon_t)$

($E(y_t) = \mu$, $E(y_{t-1}) = \mu$)

$\mu = 2 + 0.5\mu + 0$

$0.5\mu = 2$

$\mu = 4$

$Var(y_t) = 0.25 \cdot Var(y_{t-1}) + \sigma_\varepsilon^2 + 2 \cdot 0.5\, cov(y_{t-1}, \varepsilon_t)$

$cov(y_{t-k}, \varepsilon_t) = 0$ и $cov(y_t, \varepsilon_{t+k}) = 0$

$0.75\, Var(y_t) = \sigma_\varepsilon^2$

$Var(y_t) = \frac{4}{3}\sigma_\varepsilon^2$

лаги в плюс, так удобнее

$\gamma(1) = cov(y_{t+1}, y_t) = cov(2 + 0.5 y_t + \varepsilon_{t+1}, y_t) = 0.5\, cov(y_t, y_t) = 0.5\, \gamma(0)$

$\gamma(2) = cov(y_{t+2}, y_t) = cov(2 + 0.5 y_{t+1} + \varepsilon_{t+2}, y_t) = 0.5\, cov(y_{t+1}, y_t) = 0.5\ \gamma(1) = 0.5^2 \cdot \gamma(0)$

$\gamma(3) = cov(y_{t+3}, y_t) = cov(2 + 0.5 y_{t+2} + \varepsilon_{t+3}, y_t) = 0.5\, cov(y_{t+2}, y_t) = 0.5^3 \cdot \gamma(0)$

...

$\gamma(k) = 0.5^k \gamma(0) = AC$

$\rho(k) = \frac{\gamma(k)}{\sqrt{Var(y_t) \cdot Var(y_{t+k})}} = \frac{0.5^k \cdot \gamma(0)}{\gamma(0)} = 0.5^k = ACF$